PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	A systematic review of the factors and the key indicators that identify doctors at risk of complaints, malpractice claims, or impaired performance
AUTHORS	Austin, Elizabeth; Do, Vu; Nullwala, Ruqaiya; Fajardo Pulido, Diana; Hibbert, Peter; Braithwaite, Jeffrey; Arnolda, Gaston; Wiles, Louise; Theodorou, Tahlia; Tran, Yvonne; Lystad, Reidar; Hatem, Sarah; Long, Janet; Rapport, Frances; Pantle, Annette; Clay-Williams, Robyn

VERSION 1 – REVIEW

REVIEWER	Bradfield, Owen The University of Melbourne School of Population and Global Health
REVIEW RETURNED	06-Apr-2021

GENERAL COMMENTS	Having reviewed this paper previously, I can see that considerable work has gone into restructuring the results section. In particular, the risk factors for claims, complaints and impaired performance are now considered together. This is an improvement on the previous submission. However, I remain unconvinced that claims and complaints should be considered alongside impaired performance. According to the hypothesis of this paper, claims and complaints are the outcome of a number of different risk factors, such as demographic and workplace factors. However, one of these risk factors is also impaired performance. Therefore, while impaired performance is an outcome, it is also itself a risk factor for claims and complaints. The link between impaired performance and claims/complaints is not discussed. There is no clear explanation why the authors have extended the analysis beyond claims and complaints. I believe that this paper would be much stronger if impaired performance was removed from the analysis (so this paper focussed on claims/complaints), if it was considered as an exposure variable (rather than an outcome variable). For me, this remains a central methodological weakness of this paper. Relatedly, it is unclear why complaints and claims have been separated in the analysis. While it is not unreasonable to do so, it does make the results somewhat repetitive and I am not sure that there is a good reason to do this. What additional information does separating these different types of patient experiences tell us about the profile of doctors?

	I feel that the potential risk factors could be collapsed into a smaller number of categories. For example, diet seems only to be related to sleep. This could be included under fatigue? Likewise, could fatigue be included under mental health and burnout? The discussion section is a little longer than before, but again lacks detailed critical analysis of the problem of doctors' health or its solutions. With the greatest respect to the many contributing authors, I am wondering if the paper would be strengthened by collaborating with experts who specialise in medical law and/or medical regulatory research, and who can better contextualise these findings. The discussion in its current form bespeaks a general lack of understanding of the field. For example, the third paragraph of the discussion talks about "medical governance and regulatory bodies" but it is not clear what this means. They also talk about these bodies collecting information without seeming to understand what is or is not already collected. The authors correctly identify that the majority of claims are from the US. However, they seem to misunderstand the reasons for this. Culture and legislation may well explain this. However, the authors do not acknowledge that it is not possible to sue doctors in a number of countries studied and, in Australia at least, the overwhelming majority of claims settle on confidential terms. This leaves the explanation somewhat superficial and misleading. Finally, I remain of the view that the discussion does not do justice to the significant work that has been undertaken in analysing 67 articles. This is a shame because this is the first paper that I am aware of that generates such detailed results. Personally, what I take from this paper is that the issue of predicting doctors' health and claims/complaints is far more complex than many prior papers suggest and that it is highly dependent upon context and jurisdiction. In many countries, doctors cannot be sued (eg Scandinavia and New Zealand). What is the impact of this on doctors' health and complaints rates? The paper calls into question many of the assumptions that we make about the risks of claims and complaints. I don't think this paper helpfully contributes to the existing literature by simply calling for more education, training and support. This paper's strength lies in the depth of analysis but the understandings and insights being drawn from that analysis are very limited. In summary, I would only recommend acceptance of this paper only if significant reworking is undertaken (particularly to the discussion section) and if the authors carefully consider the need to include impaired performance in the results. I would recommend that the authors collaborate more widely with experts in this area to assist them to draw more meaningful and novel conclusions from this really extensive and important body of work.

REVIEWER	Domino, Karen University of Washington, Anesthesiology & Pain Medicine
REVIEW RETURNED	15-Apr-2021

GENERAL COMMENTS	The authors performed a comprehensive descriptive systematic review concerning physician risk factors for malpractice claims, complaints, or impairment. Twenty-three risk factors were identified. Male gender, older age, more years in practice, and larger numbers of patients were associated with malpractice claims and patient complaints. Factors associated with impaired

	physician performance were burnout and substance use disorder. No single factor was a strong predictor of compliants, malpractice claims, or impaired performance. The authors recommend collection of a range of workplace and demographic variables to establish risk profiles. Thanks to the authors for their modifications to improve this manuscript!

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1 Dr. Owen Bradfield, The University of Melbourne School of Population and Global Health, **Comment**	
Having reviewed this paper previously, I can see that considerable work has gone into restructuring the results section. In particular, the risk factors for claims, complaints and impaired performance are now considered together. This is an improvement on the previous submission. However, I remain unconvinced that claims and complaints should be considered alongside impaired performance. According to the hypothesis of this paper, claims and complaints are the outcome of a number of different risk factors, such as demographic and workplace factors. However, one of these risk factors is also impaired performance. Therefore, while impaired performance is an outcome, it is also itself a risk factor for claims and complaints. The link between impaired performance and claims/complaints is not discussed. There is no clear explanation why the authors have extended the analysis beyond claims and complaints. I believe that this paper would be much stronger if impaired performance was removed from the analysis (so this paper focussed on claims/complaints), if it was considered as an exposure variable (rather than an outcome variable). For me, this remains a central methodological weakness of this paper.	Thank you for your feedback, it is greatly appreciated. The purpose of the review was to identify risk factors associated with complaints, malpractice claims, and impaired performance. Complaints and claims may arise from impaired performance (impaired performance as an exposure variable). However, not all impaired performance results in, or justifies, a complaint or claim. Therefore, impaired performance needs to be considered as an outcome variable independent of formal complaints and claims. By extending the analysis beyond the formal outcomes of complaints and claims, we seek to capture insights into quality and safety-related problems that are not captured in formal processes. We have revised the introduction to provide clearer link between impaired performance and claims/ complaints. These revisions are intended to provide clearer justification of why we extended the analysis beyond formal claims and complaints to include impaired performance. See pages 4-5.
Relatedly, it is unclear why complaints and claims have been separated in the analysis.	Complaints and claims were separated in the analysis to address a previous reviewer

While it is not unreasonable to do so, it does make the results somewhat repetitive and I am not sure that there is a good reason to do this. What additional information does separating these different types of patient experiences tell us about the profile of doctors?	comment. The reviewer argued that there was too much variability in physician factors (e.g., specialty specific factors) associated with malpractice claims compared with patient complaints for them to be combined.
I feel that the potential risk factors could be collapsed into a smaller number of categories. For example, diet seems only to be related to sleep. This could be included under fatigue? Likewise, could fatigue be included under mental health and burnout?	We appreciate that 23 is a large number of key factors and that a smaller number of categories might be desirable. Given the differences in underlying mechanisms for each construct, the factors could not be collapsed. For example, fatigue may have implications for mental health and burnout, but stems from a lack of sleep or rest. Mental health and burnout may similarly have implications for quality of sleep, but stems from cognitive/ emotional processes.
The discussion section is a little longer than before, but again lacks detailed critical analysis of the problem of doctors' health or its solutions. With the greatest respect to the many contributing authors, I am wondering if the paper would be strengthened by collaborating with experts who specialise in medical law and/or medical regulatory research, and who can better contextualise these findings. The discussion in its current form bespeaks a general lack of understanding of the field. For example, the third paragraph of the discussion talks about "medical governance and regulatory bodies" but it is not clear what this means. They also talk about these bodies collecting information without seeming to understand what is or is not already collected. The authors correctly identify that the majority of claims are from the US. However, they seem to misunderstand the reasons for this. Culture and legislation may well explain this. However, the authors do not acknowledge that it is not possible to sue doctors in a number of countries studied and, in Australia at least, the overwhelming majority of claims settle on confidential terms. This leaves the explanation somewhat superficial and misleading.	The purpose of the review was to identify risk factors associated with complaints, claims and impaired performance. The review was completed in collaboration with medical, legal and regulatory experts who are part of an Australian state Medical Council. We have included an example to clarify what is meant by "medical governance and regulatory bodies (e.g., Medical Board of Australia, Federation of State Medical Boards - America)" See page 70. It is reasonable to assume that governing bodies collect information about medical practitioners. It is also reasonable to expect diversity in the amount and type of information collected by these bodies. Given the purpose of the review was to identify risk factors and that governing bodies likely differ in the amount and type of information they currently collect, it was not considered feasible to go into detail in the discussion what is or is not already collected. The following example has been added to better contextualise the risk factors identified by the review and to acknowledge that differences in settlement practices may explain differences in the rate of

	claims between countries: “The issue of risk factors and predicting complaints and claims is far more complex and context specific than a list of factors suggests. For example, in some countries it may not be possible to pursue lawsuits for compensation, rather compensation for poor outcomes maybe settled through non-legal means. Differences in settlement practices may explain differences in the rate of complaints and claims between countries.” See page 69-70.
Finally, I remain of the view that the discussion does not do justice to the significant work that has been undertaken in analysing 67 articles. This is a shame because this is the first paper that I am aware of that generates such detailed results. Personally, what I take from this paper is that the issue of predicting doctors' health and claims/complaints is far more complex than many prior papers suggest and that it is highly dependent upon context and jurisdiction. In many countries, doctors cannot be sued (eg Scandinavia and New Zealand). What is the impact of this on doctors' health and complaints rates? The paper calls into question many of the assumptions that we make about the risks of claims and complaints. I don't think this paper helpfully contributes to the existing literature by simply calling for more education, training and support. This paper's strength lies in the depth of analysis but the understandings and insights being drawn from that analysis are very limited.	We agree that the issue of risk for complaints, claims and impaired performance is far more complex than many papers suggest and likely highly dependent on personal, organisational, broader health system, legislative/ litigative system and social culture than many papers suggest. The impact of differences in litigation culture/ practice on doctors’ health and complaint rates was unfortunately beyond the scope of the current review. The review contributes to the existing literature by identifying the diversity in factors that can contribute to medical practitioners risk for complaints, claims and impaired performance. We highlight in the discussion that risk factors are likely multifactorial and interdependent and suggest that the identified factors might be considered by medical practitioners, peers and employers when developing and providing supports for medical practitioners, as well as in the development and review of organisational practices and processes. Given the volume of factors it was not feasible to provide understanding and insights for all risk factors. We have revised the discussion to provide an example of the understandings and insights that might be drawn from the analysis: “For instance, understanding that high patient load is a risk factor, medical practitioners might make strategic decisions around the volume of work they choose to take on. Similarly, employers might use the understanding that

	high patient volume puts medical practitioners and patients at risk to establish safe and effective workload policies and organisational culture practices." See page 71
In summary, I would only recommend acceptance of this paper only if significant reworking is undertaken (particularly to the discussion section) and if the authors carefully consider the need to include impaired performance in the results. I would recommend that the authors collaborate more widely with experts in this area to assist them to draw more meaningful and novel conclusions from this really extensive and important body of work.	We thank you for your consideration of our work. We have revised the manuscript to address the concerns you raised, particularly the discussion section. After discussions within our team and with experts in medical practitioner governance on the inclusion of impaired performance in the results, it was decided that impaired performance should be included for the following reasons: • Impaired performance was part of the original research question. The research question informed all the decisions we made in the systematic review process including the search strategy, inclusion/ exclusion criteria and data extraction. To remove impaired performance at this late stage would contravene the high-quality systematic review standards of practice to which the current review was performed. • Removal of impaired performance as an outcome from the current review would negate the fact that impaired performance is an outcome, hiding the fact that not all instances of poor patient experiences lead to formal complaints or claims. Therefore, identifying factors related to impaired performance as an outcome is critical for quality and safety before it results in poor patient outcomes and formal complaint or claims proceedings.
Reviewer: 2 **Dr. Karen Domino, University of Washington**	
The authors performed a comprehensive descriptive systematic review concerning physician risk factors for malpractice claims, complaints, or impairment. Twenty-three risk factors were identified. Male gender, older age, more years in practice, and larger numbers of patients were associated with malpractice claims and patient complaints. Factors associated with impaired physician performance were burnout and substance use disorder. No single factor was a strong predictor of complaints, malpractice claims, or impaired performance. The authors recommend	Thank you for your feedback, it is greatly appreciated.

collection of a range of workplace and demographic variables to establish risk profiles. Thanks to the authors for their modifications to improve this manuscript!	

VERSION 2 – REVIEW

REVIEWER	Bradfield, Owen The University of Melbourne School of Population and Global Health
REVIEW RETURNED	29-Jul-2021

GENERAL COMMENTS	Dear authors, Thank you for your further revision and detailed response to my prior concerns. As I have said previously, this is a really important and novel piece of work and I congratulate the authors for extending the field so significantly. 1. While I remain of the view that the paper would be better served by separating complaints from impaired performance, the authors have provided a persuasive and compelling justification for leaving the manuscript as is. I had not appreciated that to separate the issues now would undermine the validity of the methodology. 2. While I appreciate that it may be challenging to collapse some of the risk factors identified, it does leave me (as a reader) grappling to try to make sense of what this all means and how we can improve performance and reduce complaints. I think my earlier comments about the brevity of the discussion might relate to this. I feel that the discussion is improved and I would be happy to accept it as is. However, if the editor is minded, I would perhaps recommend a couple of extra points not only about how much more complex this field is, but that it therefore doesn't come down to simply education etc. It instead requires a holistic system wide "shift" in how we approach this area. This paper may prompt that shift. Finally, I want to sincerely thank the authors for their very hard work in resubmitting this manuscript and taking the time to respond to my many prior queries. This is a really crucial piece of work that will substantially contribute to this field. The problem remains knowing what this all means and what to do about it. I think it's okay to acknowledge that dilemma in the manuscript. I look forward to seeing this work published. Dr Owen Bradfield